

Cloud Evidence Acquisition — Pre-Engagement Client Questions

Written analysis: a framework of technical and legal questions to resolve with a client before accepting a cloud data acquisition and analysis engagement.

Scenario

A client can grant access to data hosted in the cloud but needs the investigator to perform the acquisition and analysis. Before accepting the case, the investigator needs to resolve a set of technical and legal unknowns that will shape whether the engagement is feasible and how it should be scoped.

Technical Questions

- Which provider hosts the data (e.g., AWS, Google Cloud, Azure), and under which service model (IaaS, PaaS, or SaaS)? This determines what's actually accessible: IaaS typically allows inspection of OS and network logs, while SaaS usually limits access to what the application interface exposes.
- What specific data or file types are relevant, and are backups, snapshots, or logs available to support the acquisition?
- What is the expected data volume, and are there network speed or access-window constraints that could affect collection?
- Will access to the cloud environment be direct, or provided through the client's team or the cloud provider?
- Is the data encrypted, who holds the encryption keys, and will those keys be accessible during the investigation?
- What is the incident timeline, and is any relevant data volatile or at risk of being overwritten? Cloud environments are dynamic, and this determines collection priority.

Legal Questions

- What authorization exists for this access and analysis — a signed SLA, client consent, or a court order?
- Do any third parties own or hold rights to the data? Their consent may be required, and without it there's a real risk of commingled data — inadvertently collecting other customers' private information alongside the target data.
- Where are the physical servers or data centers located? Cross-border storage can introduce jurisdictional issues, since data in another country may fall under different privacy and discovery laws.
- Does the data contain PII or other sensitive material subject to regulations such as HIPAA or GDPR?
- What procedures are already in place to document and preserve data integrity, and have any legal holds been issued to prevent deletion or modification?
- Are there ongoing or anticipated legal proceedings tied to this data that should shape how the case is handled?

Why This Matters

Resolving these questions upfront allows the investigation to be planned around real constraints — anti-forensics, distributed data, jurisdiction, and chain of custody — rather than discovering them mid-engagement. In cloud forensics specifically, evidence can be lost or altered quickly if these unknowns aren't addressed before acquisition begins.